

# From Gamebook Graphs to Narrative Experience

A profile-dependent Bag-of-Paths model enriched with local LLMs

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Guillaume Gux

University of Lausanne

Can structural differences between player profiles become perceptible differences between complete stories?

# A graph captures choices—but loses their meaning

## Paragraph 30

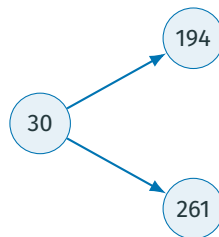
A wagon carrying small children breaks down as a Kraan approaches. The children scream in panic.

Help the children → §194

Run for cover → §261

A conventional graph records two outgoing edges, but not what distinguishes them as actions.

Example from Joe Dever, *Flight from the Dark*, §30.



Same topology  
Different narrative meaning

# Lone Wolf 1 as a controlled case study

350 narrative sections • choices, Kai disciplines, combat, inventory, endurance and chance

## Semantically modelled

### Risk

cautious / neutral / reckless

### Morality

selfish / neutral / noble

### Action

physical / neutral / tactical

## Fixed-probability assumptions

Combat win: **0.833**

Kai available: **0.5**

Escape available: **0.5**

**has\_condition: 0.5**

Values are experimental parameters,  
not player-profile dimensions.

## Not dynamically simulated

Detailed inventory state

Gold and object histories

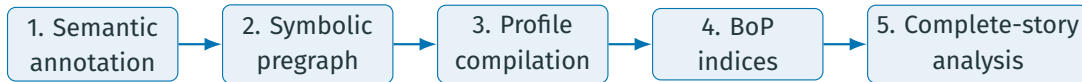
Changing endurance

Combat difficulty over time

These remain explicit limits of the  
present model.

**Modelling boundary:** the aim is a transparent experimental abstraction—not a complete game engine.

# An auditable pipeline from text to trajectories



## Why a local model?

- no dependency on a commercial API;
- exact weights and revision can be archived;
- reproducible parameters and predictable cost;
- compatible with critical DH practice.

## What makes it auditable?

- readable prompts and codebooks;
- constrained JSON outputs;
- textual evidence references;
- human supervision and calibration.

## Position

Local does not mean transparent by itself: the LLM is a documented measurement instrument, not an omniscient reader or literary judge.

# Recovering the semantics of local choices

## Input: paragraph and available choices

**Context:** children are stranded under an approaching attack.

A. “If you wish to help the children, turn to 194.”

B. “If you’d rather run for the cover of the trees, turn to 261.”

## The model returns structured evidence

transition type, three semantic axes, justification, warning status and destination—never a transition probability.

|          | Help     | Run      |
|----------|----------|----------|
| Risk     | reckless | cautious |
| Morality | noble    | selfish  |
| Action   | physical | physical |

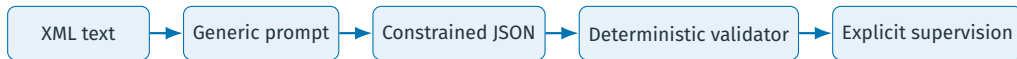
## Machine-readable output

```
transition_kind: explicit_choice
semantic_risk: reckless
semantic_morality: noble
semantic_action: physical
```

text → auditable labels

Final annotation table: LW01\_e\_edges.csv.

# Prompt refinement remains inspectable



556/556

expected choice edges  
extracted

100%

reachable from the start

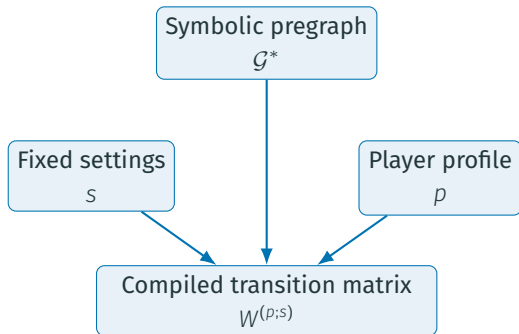
0

final schema violations

## Methodological rule

Ambiguous cases remain visible and supervised. A prompt revision must express a generic rule; it cannot memorise one paragraph to improve a small calibration set.

# One symbolic pregraph, many player profiles



One reusable pregraph is compiled into all 27 factorial profiles.

**Trajectory probability:**  $P(\pi \mid p) = \prod_{(i,j) \in \pi} w_{ij}^{(p)}$ . Profiles change the story distribution, not the text.

## Affinity and local normalisation

Each semantic axis contributes a coefficient:

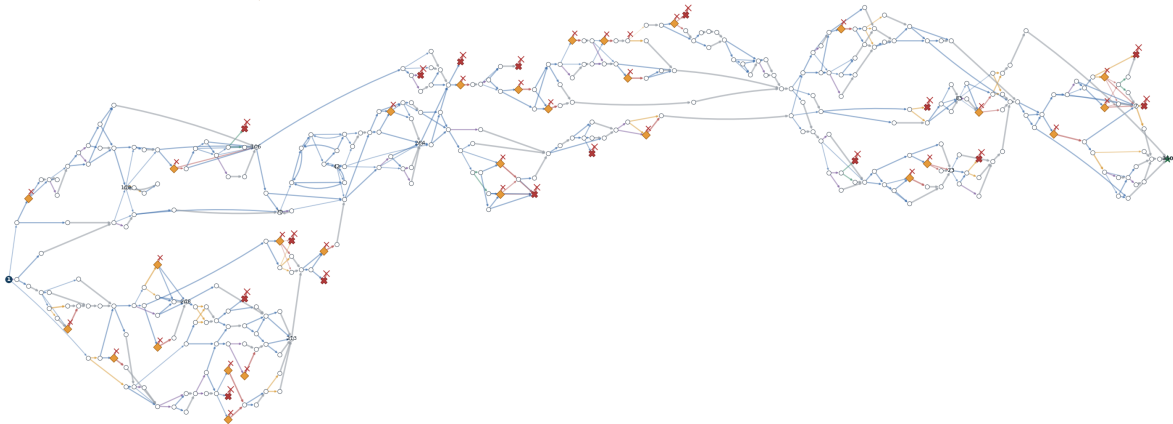
$$r_p(e) = \prod_{a \in \{r,m,c\}} \alpha_a(p, e), \quad \alpha \in \{2, 1, 0.5\}.$$

matching = 2, neutral = 1, opposed = 0.5.

$$w_{ij}^{(p)} = \frac{r_p(i, j)}{\sum_k r_p(i, k)}, \quad \sum_j w_{ij}^{(p)} = 1.$$

## LW01 - graph

Profile choice Kai discipline Combat / escape Transition to Death Death ending  
Random Condition Forced transition Combat paragraph Victory ending



Narrative projection of 350 sections. The canonical model contains 352 nodes and 602 multiedges; shared Death/Win outcomes are represented locally. Edge opacity follows the compiled weights of neutral\_neutral\_neutral. Layout coordinates: Project Aon.

# Bag-of-Paths aggregates all possible trajectories

## Random-walk BoP boundary

For the transient block  $Q$  of each compiled  $W^{(p)}$ :

$$N = (I - Q)^{-1}.$$

$N_{si}$  is the expected number of visits to paragraph  $i$  from the start  $s$ .

## Analytical, not sampled

Absorption, visits, edge flows, entropy and conditional quantities are computed in closed form for each of the 27 profiles.

## Three questions

1. **Importance:** which passages form the narrative backbone?
2. **Danger:** where does expected mortality accumulate?
3. **Sensitivity:** where and how do player profiles diverge?

### Local indices

paragraphs and edges

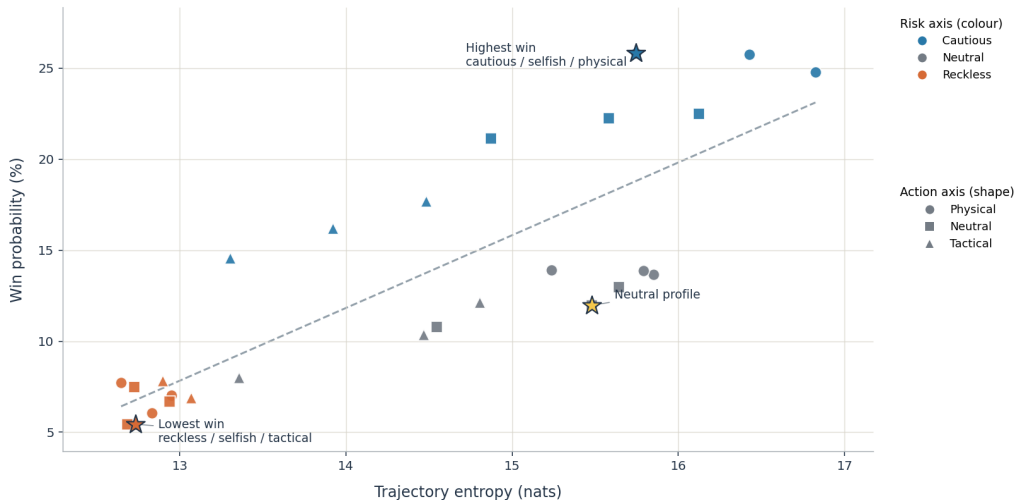
### Global indices

complete reading experience

Phase 4 uses the Random-Walk limit of the Bag-of-Paths formalism.

# LW01 — player profiles change both survival and narrative freedom

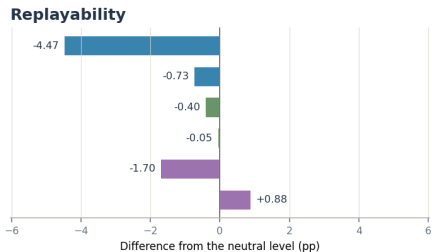
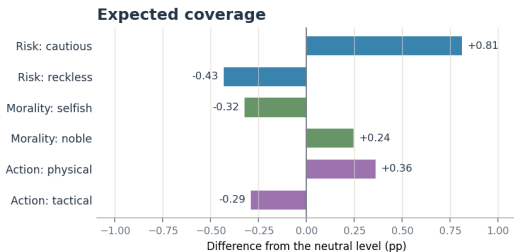
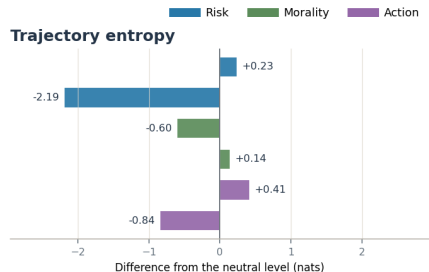
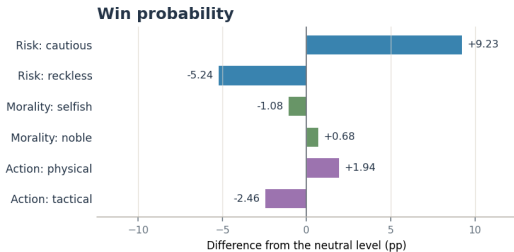
Across the 27 configured profiles, win probability and path entropy rise together (descriptive  $r = 0.83$ ).



Each point is one profile. The correlation describes the complete factorial design; it is not a population estimate or a causal effect.

# LW01 — risk is the dominant behavioural axis

Marginal effects average over the nine combinations of the other two axes.



Percentage-point differences are used for probabilities. Each comparison uses the neutral level of the same axis as its baseline.

# LW01 — three local views of the same narrative graph

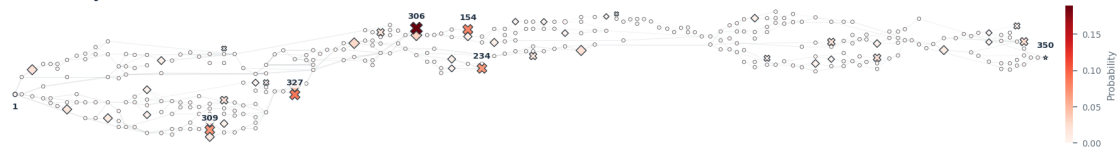
The Project Aon layout is fixed across panels; only the BoP node metric changes.

● Ordinary    ◆ Combat    ✕ Death    ★ Win    ● Start

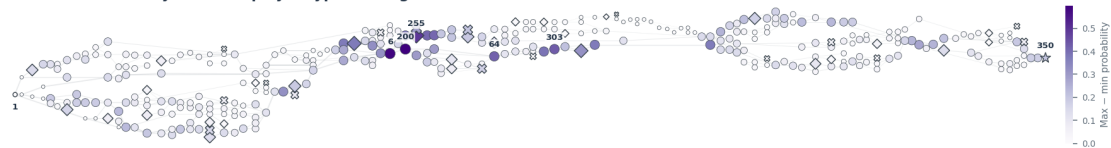
## A Neutral visit probability — the common narrative backbone



## B Mortality contribution — where neutral runs end



## C Profile sensitivity — where player types diverge



Node size and colour encode the panel metric. Faint edges are scaled by neutral expected flow. Labels identify each panel's five largest values plus \$1 and \$350.

# From structural differences to complete stories

## Controlled design

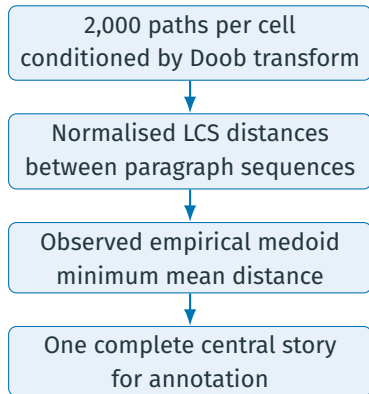
Seven profiles isolate the neutral centre and both extremes of each axis:

neutral cautious reckless  
selfish noble physical tactical

Each is crossed with **Win** and **Death**:  $7 \times 2 = 14$  cells.

## Why not the MAP path?

The most probable single path favours short stories. We need a central observed path, not the shortest likely one.



# Treating the local LLM as a calibrated instrument

## Qwen3.6-27B receives

- complete story text;
- eligible player choices;
- opaque identifiers;
- constrained output schema.

## It does not receive

generator profiles, BoP indices, phase-1 semantic labels or private outcome metadata.

| Trial | Valid | Objective finding                             |
|-------|-------|-----------------------------------------------|
| P01   | 7/7   | 14 inadmissible mechanical evidence citations |
| P02   | 5/7   | allow-list helps; two outputs quarantined     |
| P03   | 7/7   | choice/mechanics split; 0 violations          |

## Freeze before the complete run

Four individual stories and three pairs were human-annotated for prompt calibration. P03 produced **32 concordances over 44 fields**, with two human abstentions. This is **not accuracy** and not held-out validation.

P03 frozen



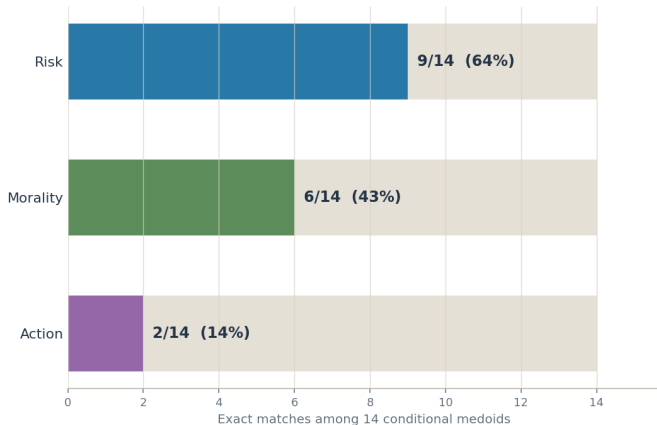
26 final tasks



0 quarantined

# LW01 — player profiles leave uneven traces in complete stories

Qwen inferred the three axes from 14 blinded conditional medoids, without profiles, edge labels or BoP indices.



**14/14**

**Causally continuous stories**

No explicit unsupported dependency was detected.

**9/14**

**Coherent perceived profiles**

The remaining 5 stories were labelled mixed.

**Why is the action score low?**

Exact recovery by generated action level

Neutral

**0/10**

Physical

**0/2**

Tactical

**2/2**

Absolute agreement includes neutral levels. It describes these selected central trajectories—not model accuracy or all possible playthroughs.

# LW01 — profile contrasts are visible, but axes co-vary

Six extreme-profile pairs were judged in both orders; B/A directions were inverted before exact stability checks.

**5/6**

**Controlled contrasts recovered**

Only order-stable controlled-axis labels count.

**18/24**

**Labels stable under A/B reversal**

Narrative distinctness plus three profile shifts.

**9/9**

**Stable off-axis labels shifted**

3 more were order-sensitive.

| Pair | Controlled design | Structural distance | Narrative A/B → B/A | Controlled contrast |
|------|-------------------|---------------------|---------------------|---------------------|
| C001 | Risk · Win        | 0.61                | High → High         | Recovered           |
| C002 | Risk · Death      | 0.79                | High → High         | Recovered           |
| C003 | Morality · Win    | 0.41                | Medium → Medium     | Recovered           |
| C004 | Morality · Death  | 0.21                | High → Medium       | Recovered           |
| C005 | Action · Win      | 0.38                | High → Medium       | Order-sensitive     |
| C006 | Action · Death    | 0.30                | Medium → Medium     | Recovered           |

Structural distance =  $1 - \text{normalized paragraph-sequence LCS}$ . Off-axis shifts show that the generated axes are not narratively orthogonal.

# A reproducible bridge between choices, structure and stories

## What we learned

1. Profiles substantially alter the probabilistic structure.
2. Their contrasts are usually perceptible in complete stories.
3. Risk, morality and action are not narratively orthogonal.

## Current limits

1. one gamebook;
2. simplified mechanics and character state;
3. small human calibration set, without held-out validation.

## Next test

Apply the same pipeline and the **frozen prompt** to LW02.

Transfer, rather than further calibration, is the next methodological test.

The contribution is not an LLM that “understands” literature, but an **auditable pipeline** connecting local choices, probabilistic structure and complete narrative experience.

Code, prompts, schemas, annotations and derived tables are versioned separately.